774-578-7186|508-579-0762
greenideainsulation@gmail.com
www.greenideainsulation.com

CLOSED CELL SPRAY FOAM

Enverge NexSeal

Medium-Density Closed Cell Spray Polyurethane Foam

Manufactured by **Holcim Solutions and Products US, LLC** · Manufactured in Spring, Texas · Country of origin: **United States**

Enverge NexSeal is a spray-applied, two-component closed cell polyurethane foam (ASTM C1029 Type II) used to insulate and seal walls, attics, ceilings, crawl spaces, ducts and interior assemblies. It delivers the highest R-value per inch of any common insulation, stiffens the structure, and at 1-5/8 inches also serves as a Class II vapor retarder — insulation, air seal and vapor control in one application.

R-7.2	>90%	2.1 lb/ft³	CDPH 01350
per inch, aged · R-28 at 4"	closed cell content	nominal density	low-emitting certified
ASTM C518	ASTM D6226	ASTM D1622	Indoor air quality

WHY WE SPECIFY IT

1 Highest R-value per inch

R-7.2 per inch aged, R-28 at 4 inches. In shallow cavities — rim joists, 2×4 walls, cathedral ceilings — closed cell reaches code R-values that open cell and batts cannot.

2 Air seal and vapor retarder in one layer

Air permeance below 0.02 L/s·m² at 75 Pa, and a Class II vapor retarder (≤ 1.0 perm) at 1-5/8 inches per ICC and ASHRAE — basements and rim joists get insulation and vapor control from the same pass.

3 Stands up to water and moisture

Water absorption of only 0.29% by volume after 96 hours under a 2-inch head, and no mold or fungal growth (ASTM C1338): the right foam against concrete, in crawl spaces and wherever condensation is a risk.

4 Dimensionally stable across extremes

Under 5% change at -20°F and under 10% at +200°F and at 158°F with 100% humidity. Rated for ambient service from -40°F to 180°F, with intermittent exposure to 220°F.

HEALTH & INDOOR AIR QUALITY

- **CDPH Standard 01350 compliant.** NexSeal is a low-VOC emitting material certified to the California Department of Public Health standard. This program applies strict certification criteria and builds in safety factors specifically to account for sensitive individuals such as children and the elderly, and certifies a product as acceptable for use in environments such as schools and healthcare facilities.
- **Referenced by CHPS and LEED.** The same 01350 certification is referenced by the Collaborative for High Performance Schools and the LEED Building Rating System as a low-emitting insulation credit.
- **No mold or fungal growth.** Tested to ASTM C1338 with no growth; closed cell foam absorbs almost no water (0.29% by volume).
- **Our standard practice with children in the home.** We recommend families with young children, elderly members, or anyone chemically sensitive plan to be out of the house during spraying and for a full 24–48 hours afterward while the foam finishes curing.

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TECHNICAL PROPERTIES

PROPERTY	VALUE	TEST METHOD
Apparent density	**2.1 lb/ft³ (±10%)**	ASTM D1622
R-Value (aged) at 75°F mean	**7.2 per inch · R-28 at 4"**	ASTM C518
Closed cell content	**> 90% by volume**	ASTM D6226
Compressive strength	**25 lb/in² nominal**	ASTM D1621
Tensile strength	**50 lb/in²**	ASTM D1623
Water absorption (96 h, 2" head)	**0.29% by volume**	ASTM D2842
Water vapor permeance at 1.6 in.	**≤ 1.0 perm**	ASTM E96
Air permeance at 75 Pa	**< 0.02 L/s·m²**	ASTM E2178
Fungi resistance	**No growth**	ASTM C1338
Dimensional stability at -20°F	**< 5% change**	ASTM D2126
Dimensional stability at +200°F	**< 10% change**	ASTM D2126
Dimensional stability, 158°F / 100% RH	**< 10% change**	ASTM D2126
Flame spread index (at 4 in.)	**Class 1 (A) — < 25**	ASTM E84
Smoke developed index (at 4 in.)	**Class 1 (A) — < 450**	ASTM E84
Service temperature range	**-40°F to 180°F**	220°F intermittent

CODE COMPLIANCE & CERTIFICATIONS

CSI master specification	07 21 19
Surface burning characteristics	ASTM E84 / NFPA 255 / UL 723 — Class 1 (A), tested at 4"
Thermal barrier	Separate from occupied space with an approved 15-minute thermal barrier, such as 1/2" gypsum board or equivalent (manufacturer's TDS)
Ignition barrier	No AC377 Appendix X assemblies are listed on the current NexSeal TDS — foam left exposed in attics or crawl spaces receives an approved intumescent coating (DC315 / FireShell), or we confirm the assembly with Enverge first
Vapor retarder	Class II (≤ 1 perm) at 1-5/8" per ICC and ASHRAE; Enverge recommends a minimum of 2"
Indoor air quality	CDPH Standard 01350 — low-emitting material; referenced by CHPS and LEED

ENVIRONMENTAL & CONTENT

VOC compliance
Low-emitting insulation, CA Section 01350

Recycled content
9.2% pre-consumer · 0% post-industrial

Rapidly renewable content
5.2% (ASTM D6886)

MANUFACTURER DOCUMENTATION

Technical Data Sheet (TDS)	envergesprayfoam.com
Safety Data Sheet (SDS)	Parts A and B, on request
Architectural specification	CSI 07 21 19

Full TDS and SDS for every product on your project are available on request, at no charge, same day.

COATING & ASSEMBLY NOTE

NexSeal must be separated from the interior of the building (occupied space) by an approved 15-minute thermal barrier, such as 1/2" gypsum board or an equivalent material, per the manufacturer and local code. The current NexSeal TDS does not list ignition barrier (AC377 Appendix X) or NFPA 286 coating assemblies; where the foam will stay exposed in an attic or crawl space, Green Idea applies an approved intumescent coating (DC315 or FireShell F10E) or confirms the assembly with Enverge before the job. Minimum installed thickness recommended by Enverge: 2".